

## SEMESTER S7

### HIGH SPEED INTEGRATED CIRCUITS

|                                        |                          |                    |                |
|----------------------------------------|--------------------------|--------------------|----------------|
| <b>Course Code</b>                     | <b>PEEVT 744</b>         | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week (L: T:P: R)</b> | 3:0:0:0                  | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                         | 3                        | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>          | PCEVT503 VLSI Technology | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. Understand the fundamental principles of high-speed circuit operation, including signal integrity and the impact of transmission lines on high-speed signals.
2. Develop the ability to design high-speed digital circuits with a focus on clocking, timing analysis, and interconnect design, while optimizing for low power consumption.
3. Gain proficiency in designing high-speed analog components such as amplifiers, comparators, and phase-locked loops, with an emphasis on performance and jitter reduction.
4. Acquire knowledge of advanced design techniques for ensuring signal integrity, high-speed memory design, and the testing and verification of high-speed integrated circuits.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Contact Hours</b> |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <p>Fundamentals of High-Speed Circuit Design</p> <p>Introduction to High-Speed ICs: Importance and applications of high-speed circuits in modern technology, Basic principles of high-speed operation.</p> <p>Signal Integrity: Understanding signal degradation, reflections, and crosstalk, Transmission lines and their impact on signal integrity, Impedance matching and termination techniques.</p> <p>High-Speed CMOS Circuits: Overview of CMOS technology for high-speed applications, Speed-power trade-offs and optimizing performance.</p> | <b>9</b>             |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2 | <p>High-Speed Digital Design Techniques</p> <p>Clocking and Timing Analysis: Clock distribution networks and clock skew, Setup and hold times, timing margins. Jitter and its impact on timing accuracy.</p> <p>Interconnect Design: On-chip interconnects: RLC effects and delay modelling, Crosstalk minimization and signal buffering, Advanced interconnects: Optical and 3D interconnects.</p> <p>Low-Power High-Speed Design: Techniques for reducing power consumption in high-speed circuits, Voltage scaling, dynamic voltage, and frequency scaling (DVFS), Power gating and multi-Vt techniques.</p>                                                                                                                                           | 9 |
| 3 | <p>High-Speed Analog Design</p> <p>High-Speed Amplifiers and Comparators: Design of wideband amplifiers: Gain-bandwidth trade-offs, Current-mode logic (CML) circuits for high-speed applications, Comparator design and optimization for speed.</p> <p>Phase-Locked Loops (PLL) and Clock Recovery Circuits: Basics of PLLs, Operation and applications, Design of PLL components: VCO, frequency dividers, phase detectors. Techniques for jitter reduction in PLLs.</p> <p>High-Speed Data Converters: Overview of ADCs and DACs in high-speed applications, Design challenges in high-speed data conversion, Techniques for enhancing the performance of data converters.</p>                                                                         | 9 |
| 4 | <p>Module 4: Advanced Topics in High-Speed IC Design</p> <p>Design for Signal Integrity: Techniques for mitigating signal integrity issues, ESD protection and layout considerations for high-speed ICs, Packaging and PCB considerations in high-speed design.</p> <p>High-Speed Memory Design: SRAM, DRAM, and non-volatile memory technologies for high-speed applications, Memory interface design and timing considerations, Emerging memory technologies: MRAM, RRAM, etc.</p> <p>Testing and Verification of High-Speed Circuits: Techniques for testing high-speed digital and analog circuits, Signal integrity testing, eye diagrams, and bit error rate (BER) testing, Design for testability (DFT) and built-in self-test (BIST) methods.</p> | 9 |

**Course Assessment Method**  
(CIE: 40 marks , ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Assignment/<br>Microproject | Internal<br>Examination-1<br>(Written) | Internal Examination-<br>2<br>(Written ) | Total |
|------------|-----------------------------|----------------------------------------|------------------------------------------|-------|
| 5          | 15                          | 10                                     | 10                                       | 40    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                                                        | Part B                                                                                                                                                                                                                                                                                                       | Total     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>Each question carries 9 marks.</li> <li>Two questions will be given from each module, out of which 1 question should be answered.</li> <li>Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b> |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                                                   | Bloom's<br>Knowledge<br>Level (KL) |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| CO1            | understand signal integrity issues in high-speed circuits and apply concepts of transmission lines and impedance matching effectively.                            | K2                                 |
| CO2            | design and optimize high-speed digital circuits, considering timing constraints, power consumption, and interconnect challenges.                                  | K3                                 |
| CO3            | design high-speed analog components, such as amplifiers and PLLs, with a focus on achieving high performance and minimizing jitter.                               | K3                                 |
| CO4            | apply advanced techniques for ensuring signal integrity, designing high-speed memory systems, and performing thorough testing and verification of high-speed ICs. | K3                                 |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table:**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 1   | 1   | 1   | 1   | 1   | 1   |     |     |     | 1    | 1    |      |
| <b>CO2</b> | 2   | 2   | 2   | 1   | 2   | 1   |     |     |     | 1    | 1    | 1    |
| <b>CO3</b> | 2   | 2   | 2   | 1   | 2   | 1   |     |     |     | 2    | 1    | 1    |
| <b>CO4</b> | 2   | 3   | 3   | 2   | 1   | 1   |     |     |     | 2    | 1    | 2    |

| <b>Text Books</b> |                                                                                   |                                                     |                                              |                         |
|-------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                                                          | <b>Name of the Author/s</b>                         | <b>Name of the Publisher</b>                 | <b>Edition and Year</b> |
| <b>1</b>          | Digital Signal Integrity: Modeling and Simulation with Interconnects and Packages | Brian Young                                         | Prentice Hall Modern Semiconductor Design    | 2000                    |
| <b>2</b>          | Design of High-Performance Microprocessor Circuits                                | Anantha Chandrakasan, William J. Bowhill, Frank Fox | Wiley-IEEE Press                             | 2000                    |
| <b>3</b>          | High Speed Digital Design: A Handbook of Black Magic                              | Howard Johnson, Martin Graham                       | Prentice Hall Modern Semiconductor Design    | 1993                    |
| <b>4</b>          | CMOS: Circuit Design, Layout, and Simulation                                      | R. Jacob Baker                                      | IEEE Press Series on Microelectronic Systems | 2019                    |

| <b>Reference Books</b> |                                                                      |                                            |                                                  |                         |
|------------------------|----------------------------------------------------------------------|--------------------------------------------|--------------------------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                                             | <b>Name of the Author/s</b>                | <b>Name of the Publisher</b>                     | <b>Edition and Year</b> |
| <b>1</b>               | High-Speed Signaling: Jitter Modeling, Analysis, and Budgeting       | Kyung Suk Oh, Xingchao Yuan                | Prentice Hall Modern Semiconductor Design Series | 2011                    |
| <b>2</b>               | Signal Integrity and Radiated Emission of High-Speed Digital Systems | Spartaco Caniggia, Francescaromana Maradei | Wiley                                            | 2008                    |
| <b>3</b>               | Advanced Signal Integrity for High-Speed Digital Designs             | Stephen H. Hall, Howard L. Heck            | Wiley                                            | 2008                    |
| <b>4</b>               | High-Speed Circuit Board Signal Integrity                            | Stephen C Thierauf                         | Artech House                                     | 2017                    |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Module No.</b>                     | <b>Link ID</b>                                                                                                              |
| <b>1</b>                              | <a href="https://nptel.ac.in/courses/108105375">https://nptel.ac.in/courses/108105375</a>                                   |
| <b>2</b>                              | <a href="https://onlinecourses.nptel.ac.in/noc24_ee134/preview">https://onlinecourses.nptel.ac.in/noc24_ee134/preview</a>   |
| <b>3</b>                              | <a href="https://archive.nptel.ac.in/courses/117/106/117106030/">https://archive.nptel.ac.in/courses/117/106/117106030/</a> |
| <b>4</b>                              | <a href="https://nptel.ac.in/courses/106103016">https://nptel.ac.in/courses/106103016</a>                                   |

## SEMESTER 7

### EMBEDDED NETWORKS

|                                            |                  |                    |                |
|--------------------------------------------|------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEEVT 746</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0: 0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PBEVT 504        | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To understand Basic Network Architectures and familiarize with network protocols

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Contact Hours</b> |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <p><b>Distributed Embedded System-</b> Basic Network Architectures: Client-Server, Peer-to-Peer, Network Topologies, Basic concepts and definitions of LAN, WLAN, WAN &amp; WPAN. OSI and ISO Models in Distributed Embedded Systems. Challenges in Designing Distributed Embedded Systems. Examples and applications of distributed embedded systems.</p> <p><b>Embedded network protocol CAN-</b> CAN identifier, different layers, bit encoding, message format, Bus length, baud rate, Arbitration mechanism, Error control, Controller Architecture, Applications, and examples.</p> | <b>7</b>             |
| <b>2</b>          | <p><b>Ethernet protocol:</b> Networking Devices-Routers, switches, hubs, and bridges, Ethernet protocol- Features, Frames, MAC, CSMA/CD -Principle, Back-off mechanisms.</p> <p><b>Internet protocol-</b> Higher level protocols -TCP and UDP, IP packet structure- IPv4 and IPv6, Wireless LAN Configurations -Single-cell and multicell, Infrastructure and AdHoc mode, IEEE 802.11 MAC protocol.</p>                                                                                                                                                                                   | <b>10</b>            |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | <p>Nomadic Access- IEEE 802.11k.</p> <p><b>Bluetooth Protocol:</b> Bluetooth Architecture, Bluetooth network topologies -Piconet and Scatternet. Bluetooth WPAN protocols- LMP and L2CAP, Bluetooth packet format-Classic packet &amp; BLE format. Polling mechanism, Establishing a connection, Power saving modes, Bluetooth applications, Overview of Mobile Ad hoc networks (MANETs) &amp; Vehicular Ad Hoc Networks (VANETs).</p>                                                               |   |
| 3 | <p><b>Sensor networking:</b> Wireless sensor networks – Introduction – Applications – Network Topology – Localization –Time Synchronization - Energy efficient MAC protocols –SMAC – Energy efficient and robust routing – Data Centric routing.</p> <p><b>Operating system basics: Functions</b> of OS, Kernel, types of operating systems. Introduction to Real-time operating systems:</p> <p>Tasks, process, threads, multiprocessing and multitasking, task scheduling, task communication.</p> | 7 |
| 4 | <p><b>Introduction to IoT and IoT networking:</b> devices vs. computers, technical building blocks, Basics of IoT networking, M2M area network, Modbus, ZigBee-Zigbee Architecture- LoRaWAN -Standardization and Alliances.</p> <p><b>Introduction to cloud computing:</b> Data Collection, storage, and computing using a Cloud Platform (Basics Only).</p>                                                                                                                                         | 7 |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Assignment/<br>Microproject | Internal Examination-<br>1<br>(Written) | Internal Examination- 2<br>(Written) | Total |
|------------|-----------------------------|-----------------------------------------|--------------------------------------|-------|
| 5          | 15                          | 10                                      | 10                                   | 40    |

### End Semester Examination Marks (ESE)

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                    | Part B                                                                                                                                                                                                                                                                | Total     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"><li>2 Questions from each module.</li><li>Total of 8 Questions, each carrying 3 marks</li></ul> <p>(8x3 = 24 marks)</p> | <ul style="list-style-type: none"><li>Each question carries 9 marks.</li><li>Two questions will be given from each module, out of which 1 question should be answered.</li><li>Each question can have a maximum of 3 sub-divisions.</li></ul> <p>(4x9 = 36 marks)</p> | <b>60</b> |

### Course Outcomes (COs)

At the end of the course, students should be able to:

| Course Outcome |                                                                                           | Bloom's Knowledge Level (KL) |
|----------------|-------------------------------------------------------------------------------------------|------------------------------|
| CO1            | Understand basic distributed embedded systems architectures, topologies, and protocols.   | K2                           |
| CO2            | Understand the architectures, and principles of operation of ethernet, IP, and Bluetooth. | K2                           |
| CO3            | Understand the concepts of sensor networking and RTOS                                     | K2                           |
| CO4            | Comprehend the concepts of IOT networking and cloud computing                             | K2                           |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 3   |     | 3   | 3   |     |     |     |     |     |      |      | 2    |
| CO2 | 3   | 2   | 3   | 3   |     |     |     |     |     |      |      | 2    |
| CO3 | 3   |     | 3   | 3   |     |     |     |     |     |      |      | 2    |
| CO4 | 3   | 2   | 3   | 3   |     |     |     |     |     |      |      | 2    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation



| <b>Text Books</b> |                                                                       |                                                     |                                    |                         |
|-------------------|-----------------------------------------------------------------------|-----------------------------------------------------|------------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                                              | <b>Name of the Author/s</b>                         | <b>Name of the Publisher</b>       | <b>Edition and Year</b> |
| <b>1</b>          | Embedded Systems Design:<br>A Unified Hardware/Software Introduction. | Frank Vahid, Tony Givargis                          | John & Wiley Publications          | 2002                    |
| <b>2</b>          | “Protocols and Architectures for Wireless Sensor Networks”            | Holger Karl and Andreas Wiilig                      | John Wiley & Sons Limited 2008     | 2008                    |
| <b>3</b>          | Operating systems Concepts                                            | Abraham Silber Chatz, Peter Baer Galvin, Greg Gagne | Wiley Publications                 |                         |
| <b>4</b>          | Internet of Things -Architectures and Design Principles               | Raj Kamal                                           | MacGraw Hill India Private Limited |                         |

| <b>Reference Books</b> |                                                                          |                                       |                                      |                         |
|------------------------|--------------------------------------------------------------------------|---------------------------------------|--------------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                                                 | <b>Name of the Author/s</b>           | <b>Name of the Publisher</b>         | <b>Edition and Year</b> |
| <b>1</b>               | Embedded Ethernet and Internet Complete                                  | Jan Axelson                           | Penram Publications, 2003.           | 2003                    |
| <b>2</b>               | “A Survey on Sensor Networks”,                                           | I.F. Akyildiz and Weillian            | IEEE Communication Magazine          | Aug 2007                |
| <b>3</b>               | Embedded systems an integrated approach                                  | Lyla B Das                            | pearson                              | Ist edn                 |
| <b>3</b>               | IoT fundamentals- networking technologies, protocol, and use cases.      | David Hanes, Gonzalo Salgueiro et al. | Cisco Press                          | 2017                    |
| <b>4</b>               | Computers as Components: Principles of Embedded Computing System Design. | Wayne Wolf                            | Morgan Kaufman Publishers - Elsevier | 3ed 2008                |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                                                                                 |
| 1                              | <a href="https://archive.nptel.ac.in/courses/106/105/106105193/">https://archive.nptel.ac.in/courses/106/105/106105193/</a>                                             |
| 2                              | <a href="https://nptel.ac.in/courses/108105057">https://nptel.ac.in/courses/108105057</a>                                                                               |
| 3                              | <a href="https://onlinecourses.nptel.ac.in/noc24_cs24/preview">https://onlinecourses.nptel.ac.in/noc24_cs24/preview</a>                                                 |
| 4                              | <a href="https://youtube.com/playlist?list=PLDD18E950D85E018C&amp;si=6wnLOHy0GarETWi1">https://youtube.com/playlist?list=PLDD18E950D85E018C&amp;si=6wnLOHy0GarETWi1</a> |